Casey Barton

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PROFILE

I build the systems underneath agent deployments: the tool-calling runtimes, API integrations, structured-output contracts, evaluation harnesses, and trace observability that turn one-off field wins into reusable product. I own the full loop — prototype, validate with evals, deploy to staging and production, monitor traces, and iterate on production evidence. Every agent I ship runs under an eval discipline first: golden sets, behavioral assertions, LLM-as-a-judge with written rubrics for safety, effectiveness, and task quality — and every production failure mode I have found has something built to catch it. Fail-closed by default.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops, retrieval, memory, guardrails, and typed schema validation on every tool contract; **65** MCP connectors exposing governed context to any MCP-compatible system; full run traces so any execution can be reconstructed; per-task latency and token-cost instrumentation that drives prompt and configuration optimization.

CORE COMPETENCIES

Engineering: Python (primary) · TypeScript · Node.js · SQL · Bash · REST APIs · concurrent Python I/O · HTTP APIs · webhooks · streaming workflows · containers · GitHub Actions (CI/CD) · Postgres · event-driven systems

AI & Evaluation: agent runtimes · hand-rolled multi-agent frameworks · MCP servers & connectors · LLM evaluation harnesses · quality scoring · LLM-as-a-judge with written rubrics (safety, effectiveness, quality) · golden datasets · behavioral assertions · trace intelligence · parallel tool orchestration · failure-mode instrumentation · benchmarking · function calling · structured outputs · guardrails · fail-closed recovery

Production: prototype → evals → staging/prod deploy → trace monitoring → iterate · JSON Schema / Pydantic-style validation · token budgets · retry logic · observability & run logging · latency & token-cost benchmarks · CI regression gates · documentation engineers actually read

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Own the full agent lifecycle: prototype → evals → staging/prod deploy → trace monitoring → iterate on production evidence. Prompt and configuration changes are validated by eval harnesses before they touch production.
- Design evaluation harnesses and quality scoring: golden datasets, behavioral assertions, and LLM-as-a-judge scoring with written rubrics covering safety, effectiveness, and task quality — with CI regression gates that block a regression from shipping.
- Build and ship tool-calling LLM systems: structured outputs, parallel API orchestration, typed schema validation on every tool contract (JSON Schema / Pydantic-style), bounded retries, and token budgets.

- Author **65** MCP connectors and integrations — the tools agents use: API integrations with strict validation, structured resource submission, and evidence boundaries with audit receipts.
- Design agent systems for failure: instrument the failure modes found in production — stuck loops, bad retrievals, tool misuse, context blowouts — and build the catchers: replay loops, fail-closed gates, behavioral assertions, and recovery restricted to evidence declared before reassessment.
- Drive quality / latency / cost tradeoffs with data: per-task latency and token-cost instrumentation on every run; model selection per task complexity.
- Run an evidence-grade agent control plane across a **1,395-repository** estate: guardrails, typed handoff contracts, human approval gates on consequential actions — backed by a **989-test** suite, **62/62** verification checks green.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models and fail-closed **GO / NO-GO** decisions; recovery restricted to evidence declared before reassessment.
- Verification gates on every agent action: no agent completion claim without independent provider readback — the eval gate decides, not the agent.

GlacierEQ estate — production agent systems

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Evaluation: agent runtimes · hand-rolled multi-agent frameworks · MCP servers & connectors · eval harnesses · quality scoring · LLM-as-a-judge · golden datasets · trace intelligence · parallel tool orchestration · function calling · structured outputs · memory systems · guardrails · fail-closed recovery

Production: full lifecycle ownership · JSON Schema / Pydantic-style validation · HTTP APIs & webhooks · token budgets · retry logic · CI regression gates · observability & run logging · latency/cost tradeoff analysis · cross-layer debugging · documentation & handover

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.